

FINA0064-1: Financial Risk Modeling

Project Report

Authors :
Abdoul Latif Belem s2504109
Alexandre Dutailly s212833
Louis HOGGE s192814
Thomas Rassart s210507

Professor : J. Hambuckers
Assistant : P. Hubner
Year : 2025-2026

Contents

1 Introduction **1**

2 Data Features **2**
2.1 Leptokurtosis (fat tails) 2
2.2 Volatility Clustering (heteroscedasticity) 4
2.3 Dependence structure 6
2.4 Implications for modelling 7

3 Risk analysis **7**
3.1 Marginal Volatility (GARCH) 7
3.2 Dependence Modelling (Copulas) 8

4 Forecasting methodologies **9**
4.1 Benchmark approach: RiskMetrics 9
4.2 Semi-Parametric approach: Filtered Historical Simulation (FHS) 9
4.3 Fully parametric approach: Copula-GARCH Monte Carlo 10

5 Out-of-Sample Performance (Backtesting) **11**
5.1 Backtesting framework 11
5.2 Statistical validation (Kupiec test) 11
5.3 Results discussion 12

6 Conclusion **13**

1 Introduction

The objective of this project is to measure market risk for our portfolio, evaluating the 95% Value-at-Risk (VaR) and Expected Shortfall (ES). The composition of our portfolio, as of November 6th, 2025, is available in the table 1.

Ticker	Company name	Amount	Price	Market value
JPM	JPMorgan Chase & Co.	641	$311.68	$199,786.88
TM	Toyota Motor Corp.	1439	$199.15	$286,576.85
NVO	Novo Nordisk	3423	$48.46	$165,878.58
ZBRA	Zebra Technologies Corp.	620	$266.27	$165,087.40
ASMLF	ASML Holding NV	158	$1,033.05	$163,221.90
RNMBF	Rheinmetall AG	81	$2,024.00	$163,944.00

Table 1: Portfolio composition as of November 6th

To evaluate the risk of the portfolio, we will compare three different Value-at-Risk and Expected Shortfall modelling frameworks: a parametric RiskMetrics benchmark, a semi-parametric Filtered Historical Simulation approach, and a fully parametric Copula-GARCH Monte Carlo model. Those three models are important to compare as they all capture heavy tails, volatility dynamics and cross-asset dependence differently. We will select the best model, based on both statistical criteria and economic relevance.

To conduct this analysis, we divided our dataset into two distinct periods:

- **In-Sample (Calibration):** We retrieve historical prices starting from January 1st, 2020; however, after aligning all assets on common trading dates (inner join) and computing log-returns, the effective calibration window used in the estimations runs from October 28th, 2020 to November 6th, 2025.
- **Out-of-Sample (Forecast):** We performed a fixed-origin, multi-step ahead forecast for the window from November 7th to December 7th, 2025. Parameters were estimated using data up to November 6th, and conditional volatility was projected forward for the 21-day horizon without re-estimation. This tests the model's persistence and ability to forecast volatility trends.

2 Data Features

The objective of this section is to analyse the statistical properties of the portfolio's log-returns. We will test for deviations from the normal distribution and the presence of time-varying volatility.

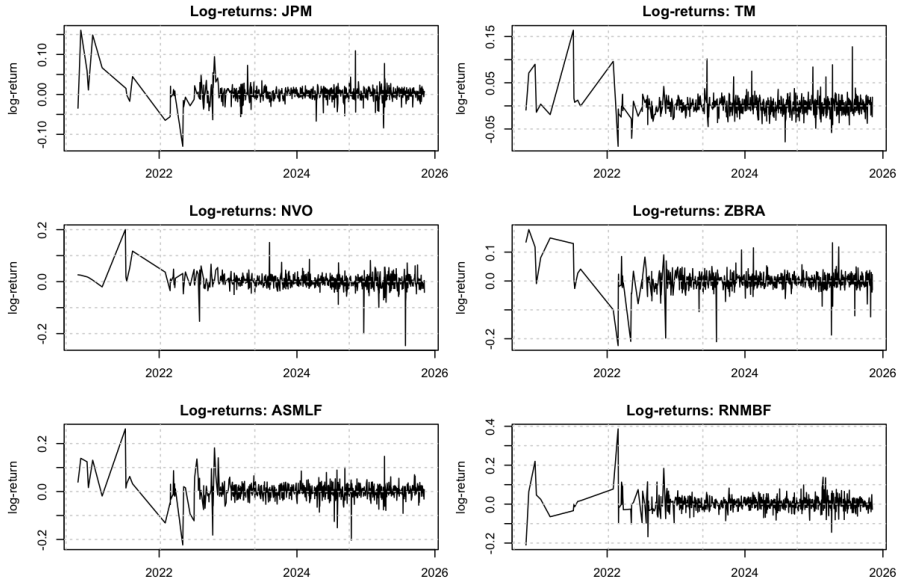


Figure 1: Time Series of Log-Returns

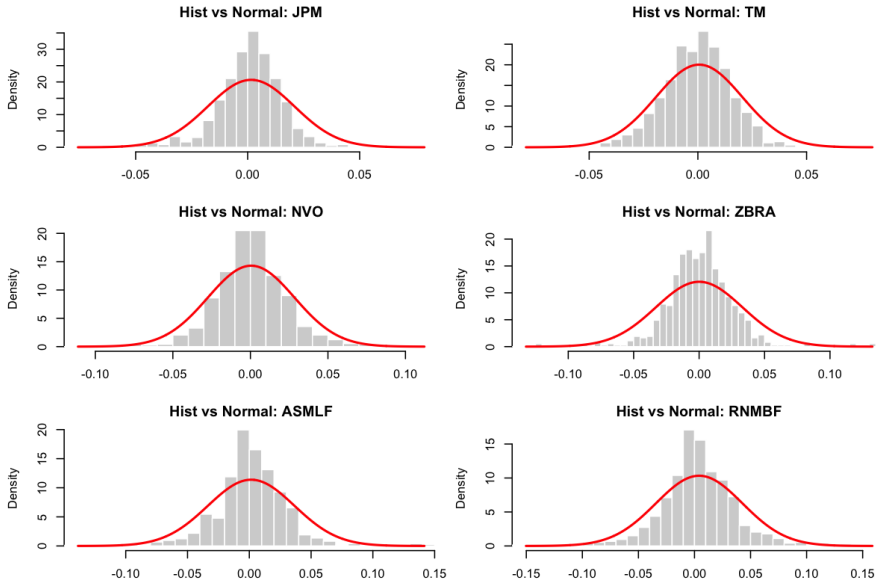


Figure 2: Empirical distribution vs Normal distribution for log-returns

2.1 Leptokurtosis (fat tails)

The Kurtosis of 13.36 (excess kurtosis = 10.36) confirms that returns are not normally distributed, and exhibit heavy tails. The deviation from normality is visually confirmed by the Q-Q plots in Figure 3, where sample quantiles deviate significantly from the theoretical normal line at the tails.

Due to this high Kurtosis, we will use the Student-t distribution, adding a degrees-of-freedom parameter to control tail thickness. Ignoring this degree of leptokurtosis would lead to a systematic underestimation of extreme portfolio losses, biasing the VaR and ES measures downward in periods of market stress.

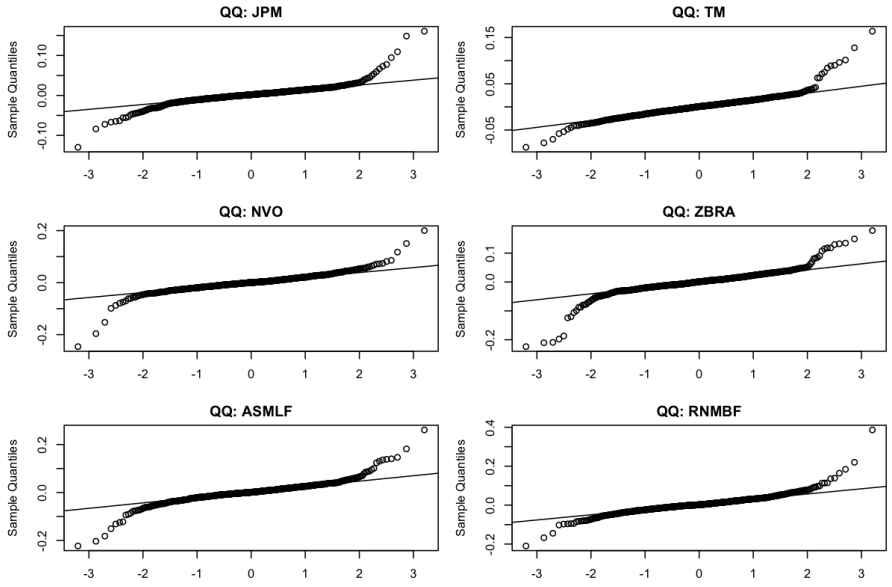


Figure 3: QQ plots of our stocks

Ticker	Mean	Sd	Skewness	Kurtosis
JPM	0.0016	0.0193	0.9802	19.5588
TM	0.0006	0.0199	1.5224	14.5712
NVO	0.0005	0.0279	-0.8030	20.1160
ZBRA	0.0000	0.0331	-1.0698	15.8164
ASMLF	0.0015	0.0350	0.0273	14.3522
RNMBF	0.0042	0.0386	1.4485	19.8567
PORT	**0.0013**	**0.0167**	**0.9180**	**13.3637**

Table 2: Descriptive Statistics of Asset Returns

While kurtosis provides a global measure of tail heaviness, it does not convey the magnitude of extreme outcomes directly. To complement this analysis, Table 3 reports empirical tail quantiles of daily log-returns at the 1%, 5%, 95%, and 99% levels for each asset and for the portfolio.

Several assets exhibit substantial tail losses at the 1% level, notably ZBRA, ASMLF, and RNMBF, confirming the presence of pronounced downside risk. The significant skewness observed in Table 2, combined with the difference between lower and upper tail magnitudes in Table 3, supports the evidence of non-Gaussian behavior and motivates the use of heavy-tailed distributions in subsequent modelling.

Ticker	1%	5%	95%	99%
JPM	-0.0549	-0.0264	0.0237	0.0578
TM	-0.0409	-0.0272	0.0256	0.0739
NVO	-0.0675	-0.0356	0.0399	0.0704
ZBRA	-0.1041	-0.0389	0.0387	0.1126
ASMLF	-0.0929	-0.0467	0.0436	0.1167
RNMBF	-0.0914	-0.0491	0.0598	0.1133
PORT	**-0.0403**	**-0.0213**	**0.0236**	**0.0490**

Table 3: Tail quantiles of daily log-returns

2.2 Volatility Clustering (heteroscedasticity)

The p-values of the Ljung-Box test are highly significant, except for NVO, which shows weaker evidence of volatility clustering. This indicates the presence of strong serial correlation in the conditional variance, confirming the stylised fact of volatility clustering. The test leads to the rejection of the null hypothesis H_0, providing evidence that periods of high market turbulence tend to persist, as large price changes are likely to be followed by further large changes.

This motivates us to use GARCH models, in order to capture conditional heteroscedasticity, as stationary variance models cannot account for the time-varying risk observed in our data. Failing to account for volatility clustering would result in risk forecasts that react too slowly to changing market conditions, leading to misestimation of portfolio risk, mainly during periods of high uncertainty.

Ticker	LjungBox_pvalue_sqLag10
JPM	0
TM	1.3457×10^{-10}
NVO	0.6470
ZBRA	2.3234×10^{-10}
ASMLF	2.6423×10^{-13}
RNMBF	3.4062×10^{-13}
PORT	**0**

Table 4: Ljung-Box Test p-values (Squared Returns, Lag 10)

Beyond formal Ljung–Box tests, volatility clustering can be visualized through the autocorrelation function (ACF) of squared returns. Figure 4 displays the ACF of squared log-returns for each asset, while Figure 5 reports the portfolio returns together with the ACF of squared portfolio returns.

Significant and persistent autocorrelation in squared returns provides clear evidence of volatility clustering, a stylized fact of financial time series that cannot be captured by homoscedastic models. These visual diagnostics further justify the use of GARCH-type models to describe conditional volatility dynamics.

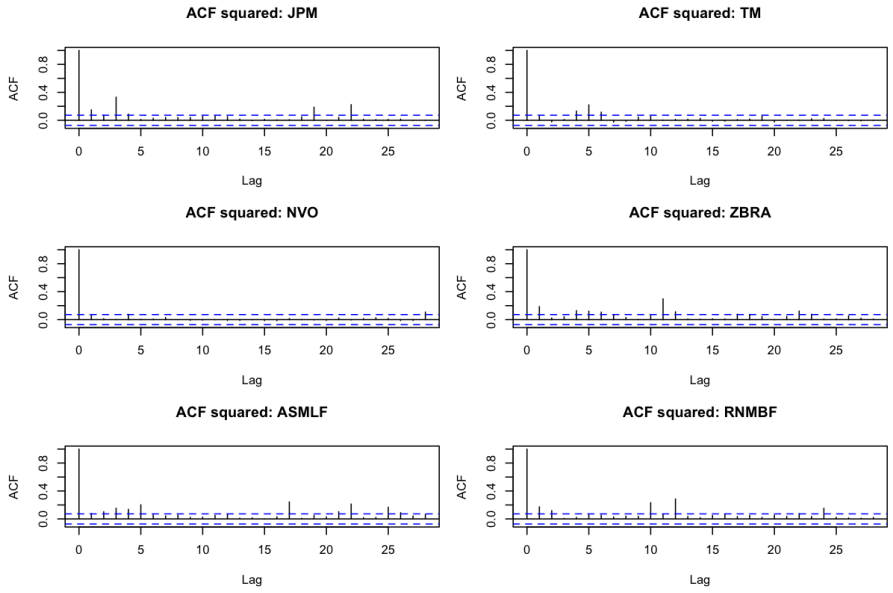


Figure 4: ACF of the squared log-returns

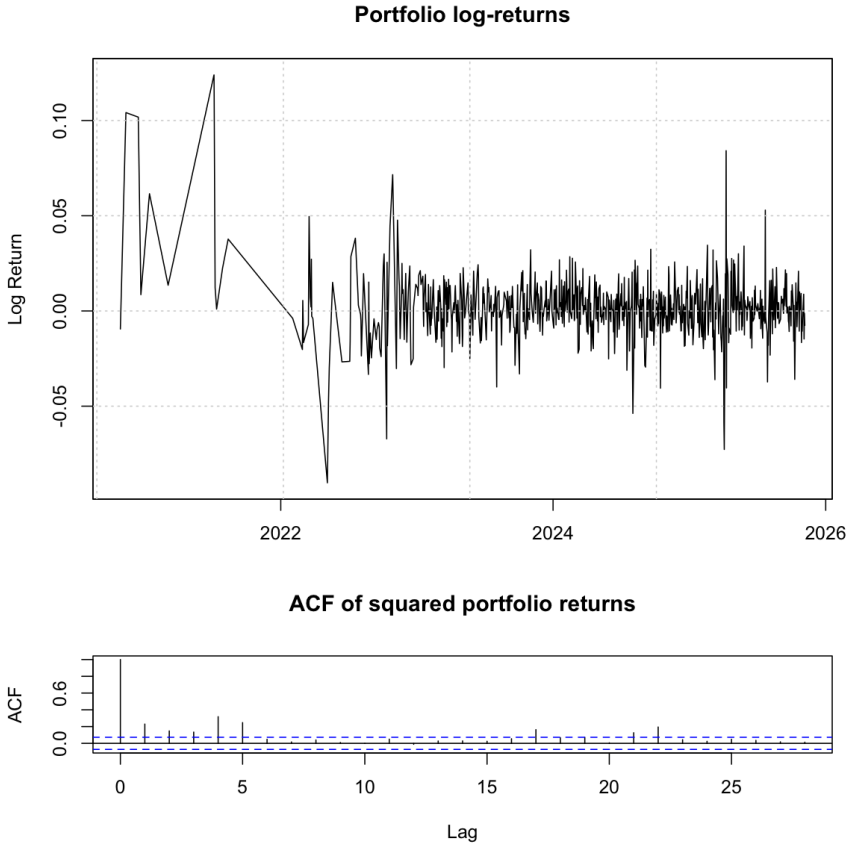


Figure 5: Portfolio log-returns and ACF of the portfolio's squared log-returns

2.3 Dependence structure

We computed the linear correlation matrix, but we know that this matrix has limitations: it is not able to capture tail dependence and non-linear co-movements (see Table 5).

	JPM	TM	NVO	ZBRA	ASMLF	RNMBF
JPM	1.0000	0.3136	0.1317	0.5035	0.4225	0.0710
TM	0.3136	1.0000	0.2563	0.3786	0.3897	0.1102
NVO	0.1317	0.2563	1.0000	0.1908	0.2564	0.0147
ZBRA	0.5035	0.3786	0.1908	1.0000	0.5245	-0.0424
ASMLF	0.4225	0.3897	0.2564	0.5245	1.0000	0.1107
RNMBF	0.0710	0.1102	0.0147	-0.0424	0.1107	1.0000

Table 5: Pearson Linear Correlation Matrix

Therefore, we need to use the Rank Correlation (Kendall's Tau) in Table 6, and Copulas, which allow modelling dependence structures independently of marginal distribution. Therefore, relying only on linear correlation would overstate diversification benefits and underestimate portfolio losses during extreme events.

	JPM	TM	NVO	ZBRA	ASMLF	RNMBF
JPM	1.0000	0.2066	0.1116	0.3066	0.1726	0.0815
TM	0.2066	1.0000	0.1507	0.2534	0.2493	0.1064
NVO	0.1116	0.1507	1.0000	0.1390	0.1738	0.0440
ZBRA	0.3066	0.2534	0.1390	1.0000	0.3224	0.0687
ASMLF	0.1726	0.2493	0.1738	0.3224	1.0000	0.0958
RNMBF	0.0815	0.1064	0.0440	0.0687	0.0958	1.0000

Table 6: Kendall's Tau Rank Correlation Matrix

Figure 6 provides a comprehensive visualization of the dependence structure across assets through pairwise scatterplots, marginal distributions, and correlation coefficients. While linear correlations remain moderate, several asset pairs exhibit visible co-movements and non-linear dependence patterns.

This joint visualization highlights the limitations of simple multivariate Gaussian assumptions and motivates the use of copula-based models capable of capturing richer dependence structures, particularly in the tails.

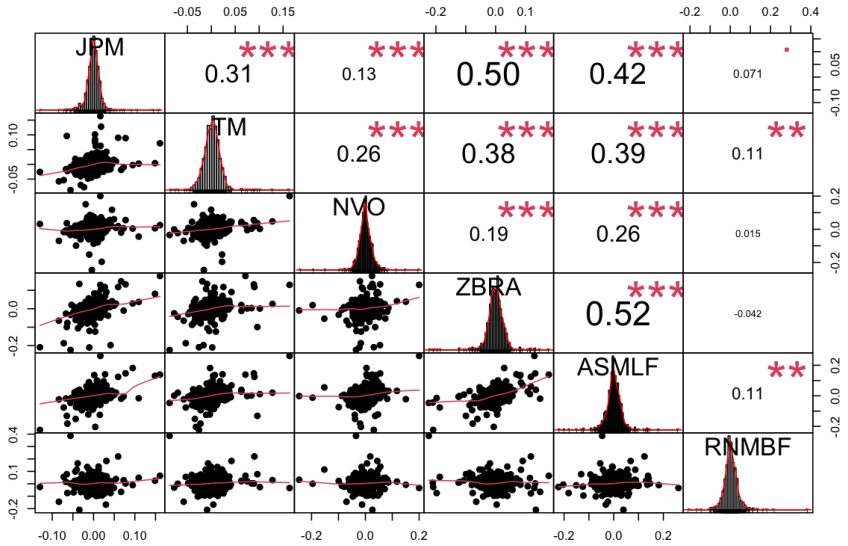


Figure 6: Correlation between assets

2.4 Implications for modelling

The analysis of returns showed deviations from the underlying static Gaussian risk models. The presence of strong Kurtosis shows that extreme losses occur more frequently than predicted by a normal distribution.

Moreover, the significant Ljung-Box test results confirm the existence of volatility clustering, implying that market risk evolves over time. Therefore, GARCH models are needed to capture time-varying risk levels.

Finally, the risk dependence analysis highlights cross-asset co-movements. Thus linear correlation measures are insufficient as they fail to capture non-linear and tail dependence. Copula-based models allow the distribution of returns to be modelled independently, and are well suited to represent the risk of simultaneous extreme losses.

Taken together, those stylised facts motivate a Copula-GARCH model, accounting for heavy tails, volatility dynamics, and multivariate dependence.

3 Risk analysis

The objective of this section is to describe the three approaches we had, and justify the choice of our final model.

3.1 Marginal Volatility (GARCH)

We compared three different GARCH models: sGARCH with normal innovations, sGARCH with Student-t innovations, and eGARCH with Student-t innovations. To compare, we used the Bayesian Information Criteria (BIC), and decided to select the Student-t based GARCH model.

Ticker	GARCH_N	GARCH_t	eGARCH_t	Best Model
JPM	-5.2992	-5.5975	-5.5902	**GARCH_t**
TM	-5.0424	-5.2558	-5.2463	**GARCH_t**
NVO	-4.3473	-4.6733	-4.6738	**eGARCH_t**
ZBRA	-4.1538	-4.4838	-4.4797	**GARCH_t**
ASMLF	-4.0777	-4.2749	-4.2611	**GARCH_t**
RNMBF	-3.9482	-4.0513	-4.0281	**GARCH_t**

Table 7: GARCH Model Comparison Using BIC

As we mentioned in the section 2.4, models assuming Gaussian innovations underperform, as they fail to capture the heavy-tailed nature of asset returns.

As shown in Table 7, the standard sGARCH model with Student-t innovations yields the best (lowest) BIC for the majority of assets. We note that for NVO, the eGARCH model provided a marginally better fit (BIC -4.6738 vs -4.6732). The eGARCH is theoretically valuable for equities as it accounts for the leverage effect, where negative returns tend to increase future volatility more than positive returns of the same magnitude. This suggests that asymmetric news impact is a relevant risk factor for NVO. However, to maintain model parsimony and consistency for the multivariate simulation, we selected the sGARCH-t specification for all assets in the portfolio.

3.2 Dependence Modelling (Copulas)

We fitted three Copula families to the standardised residuals: normal, Student-t, and Clayton. Again, we selected the best model based on the BIC, and we chose the Student-t Copula. This selection is important as the t-Copula exhibits symmetric tail dependence ($\lambda_L = \lambda_U > 0$). This captures the risk that assets crash together.

Copula	BIC
Normal	-495.0413
t	**-550.5217**
Clayton	-332.9304

Table 8: Copula Model Selection Based on BIC

The Student-t copula was chosen over the Clayton copula because it captures symmetric tail dependence. While the Clayton copula only accounts for lower-tail dependence, the Student-t copula recognizes that extreme co-movements can occur in both the upper and lower tails. This is important for our portfolio, as it ensures we do not underestimate the risk of simultaneous extreme losses during periods of global market stress.

To assess the adequacy of the selected copula, Figure 7 compares the observed probability integral transforms (PITs) with those simulated from the fitted copula for the asset pair exhibiting the strongest dependence. The close alignment between observed and simulated joint behavior indicates that the selected copula reproduces the empirical dependence structure well, providing a visual validation of the model choice.

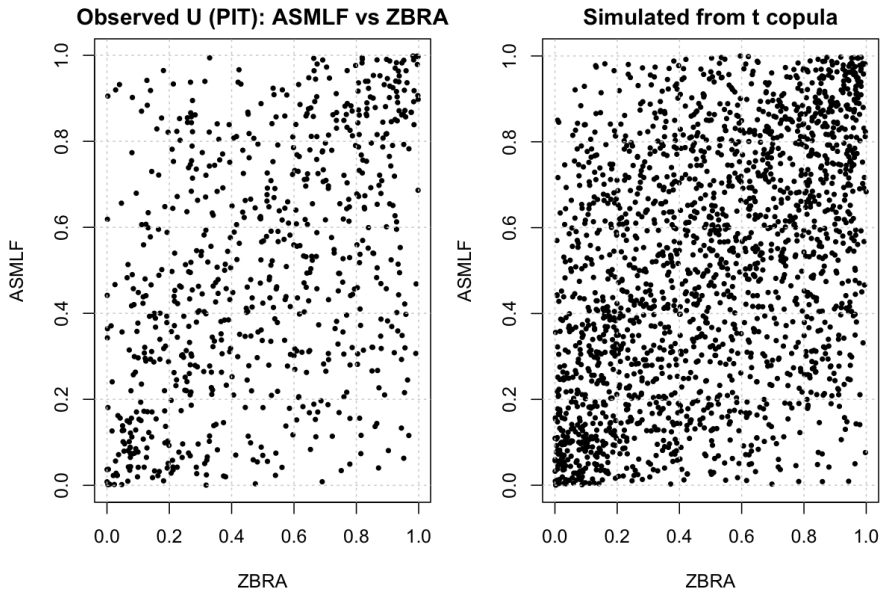


Figure 7: Copula observed vs. simulated best pair

4 Forecasting methodologies

In this section, we will detail the three risk modelling frameworks to estimate the VaR and ES for the out-of-sample period. These approaches are used to contrast the performance of parametric, semi-parametric, and simulation-based techniques. Specifically, we want to see their ability to handle the stylised facts we identified in Section 2.

4.1 Benchmark approach: RiskMetrics

We use the Variance-Covariance Method (VCV) approach as a baseline for performance comparison. Consistent with the terminology used in the course exercises, we refer to this as our **RiskMetrics benchmark**.

In our implementation, the mean vector and covariance matrix are estimated unconditionally from the in-sample returns using the sample mean and sample covariance. This baseline therefore produces constant VaR and ES forecasts over the out-of-sample horizon (i.e., no EWMA or time-varying volatility).

The main limitation of this approach is the assumption of Gaussian tails and linear dependence, which can lead to an underestimation of extreme losses when returns exhibit heavy tails and tail dependence

4.2 Semi-Parametric approach: Filtered Historical Simulation (FHS)

To address the limitations mentioned, we implemented FHS. This method combines parametric volatility filtering with non-parametric bootstrapping.

Filtering: we fit the selected sGARCH(1,1)-Student-t model to the historical returns of each asset to estimate the conditional volatility. Then, we extract the standardised residuals.

Bootstrap: for the forecast horizon $t+1$, we randomly sample vectors of standardised residuals from the historical pool. These sampled residuals are then scaled by the forecasted conditional volatility and

shifted by the conditional mean forecast, generating simulated returns of the form:

$$r_{t+1}^{(b)} = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \cdot z^{(b)}$$

Justification: this approach captures the volatility clustering via the GARCH filter, while keeping the empirical tail shape of the residuals. This accommodates non-normality, without imposing a theoretical distribution. For each forecast day, VaR and ES are estimated from B = 10,000 simulated portfolio losses.

4.3 Fully parametric approach: Copula-GARCH Monte Carlo

Among the three models, Copula-GARCH Monte Carlo (CopulaMC_t) is our chosen model for VaR and ES for Nov 7–Dec 7. It is a fully parametric Monte Carlo simulation that explicitly models the dependence structure using Copula theory. This approach relies on Sklar's theorem, which allows the joint distribution F to be decomposed into marginal distributions and a Copula.

$$F(x_1, ..., x_d) = C(F_1(x_1), ..., F_d(x_d))$$

We forecast the conditional mean and volatility for each asset using the sGARCH(1,1)-Student-t model validated in the previous section. The Student-t shape (degrees-of-freedom) parameter is estimated in-sample for each asset and kept fixed during simulation. We simulate 10,000 draws from the selected Copula, and transform the simulated copula values into asset returns, using the inverse CDF of the standardised Student-t marginals. A key advantage of this approach over FHS is its ability to extrapolate: by sampling from a theoretical distribution rather than bootstrapping finite historical data, we can capture extreme joint-tail events that are statistically probable given the tail dependence ($\lambda > 0$), even if they were not observed in the historical window.

The figure 8 illustrates the daily 95% VaR forecasts, generated by the three approaches we described here above. We clearly see the static RiskMetrics estimates, in comparison with the FHS and Copula-MC, reacting to changes in volatility conditions.

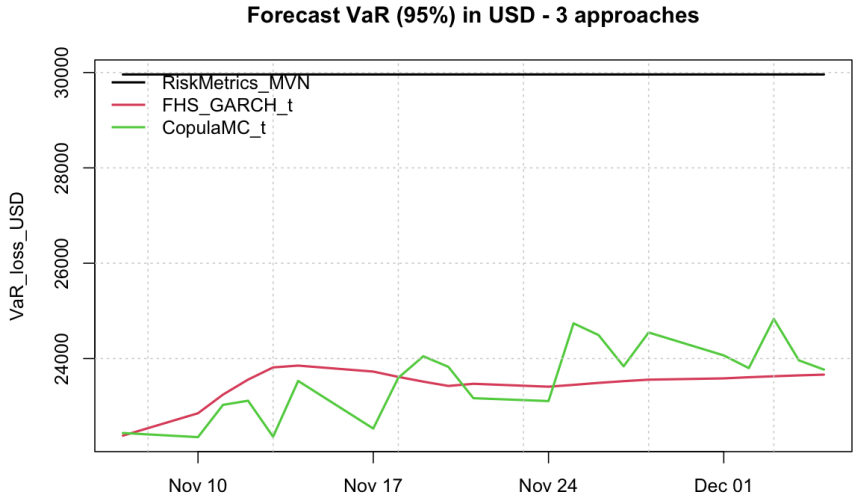


Figure 8: Forecasted 95% VaR

Additionally, as shown in figure 9, the ES estimates are consistently higher in magnitude than the VaR figures. This confirms that ES captures the severity of potential extreme losses lying beyond the 95% threshold.

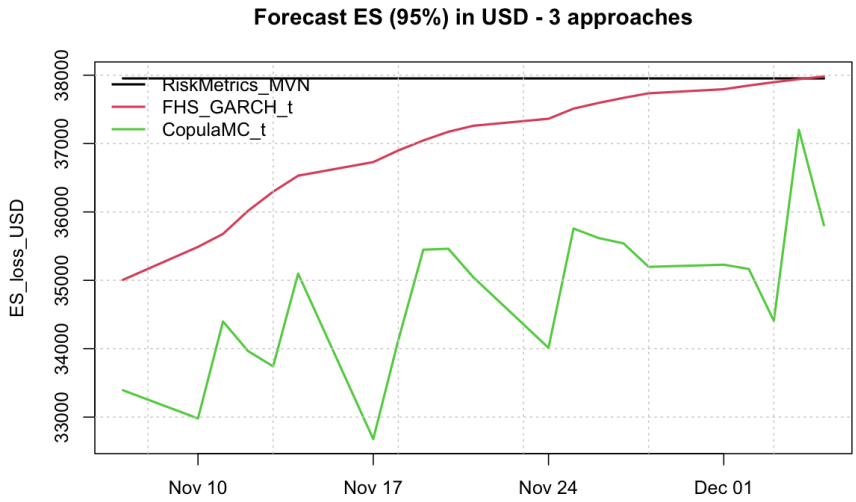


Figure 9: Forecasted ES

To summarize the empirical motivations behind the selected modeling framework, Table 9 reports a set of simple diagnostic indicators derived from the data. The presence of high average excess kurtosis, widespread volatility clustering, and non-negligible cross-asset dependence jointly support the adoption of a Copula–GARCH Monte Carlo approach for risk forecasting.

Diagnostic signal	**Value**
Average excess kurtosis (assets)	14.38
Fraction of assets with Ljung-Box $p < 0.05$	0.83
Average absolute Pearson correlation	0.25

Table 9: Summary diagnostics motivating the Copula-GARCH Monte Carlo framework

5 Out-of-Sample Performance (Backtesting)

To validate the accuracy of our risk measures, we did a backtesting over the period from November 7 to December 7 (21 observations). We evaluate the models by comparing our forecasted values with realised portfolio losses.

5.1 Backtesting framework

We define an exception (hit indicator) for each day:

$$I_t = \mathbf{1}\{L_t > \text{VaR}_t(\alpha)\}$$

where $L_t = -r_t$ is the realised loss, i.e. the negative daily log-return of the portfolio. For a confidence level of 95%, the expected hit ratio is 5%.

5.2 Statistical validation (Kupiec test)

To test our risk measures, we use the Unconditional Coverage (UC) Test. The null hypothesis H_0 : $p = \alpha$ states that the probability of failure equals the expected level. The likelihood ratio statistic is given

by:

$$LR_{UC} = -2\ln\left[(1-\alpha)^{N-x}\alpha^{x}\right] + 2\ln\left[(1-\hat{p})^{N-x}\hat{p}^{x}\right]$$

where x is the number of exceptions, and N is the sample size. Also, under H_0, $LR_{UC} \sim \chi^2(1)$.

In addition to unconditional coverage, we also report Christoffersen's tests. The Independence (IND) test assesses whether VaR violations are independent over time (i.e., no clustering of exceptions), and the Conditional Coverage (CC) test jointly evaluates correct exception frequency and independence:

$$LR_{CC} = LR_{UC} + LR_{IND}, \qquad LR_{IND} \sim \chi^2(1), \qquad LR_{CC} \sim \chi^2(2).$$

Given our short backtesting window (21 days), these tests have limited power, but they provide a useful diagnostic beyond UC.

5.3 Results discussion

Table 10 summarises the backtesting performance. It is important to note that our out-of-sample period consists of only 21 business days. Consequently, the statistical power of the Kupiec test is limited.

Model	**Exceptions**	**Hit Ratio**	p_{UC}	p_{IND}	p_{CC}
RiskMetrics_MVN	0	0.0000	0.1422	1.0000	0.3406
FHS_GARCH_t	0	0.0000	0.1422	1.0000	0.3406
CopulaMC_t	1	0.0476	0.9598	0.7427	0.9463

Table 10: VaR backtesting summary

We see that both RiskMetrics and FHS produce 0 exceptions. The p_{UC} of 0.1422 indicates that we cannot statistically reject the models. The lack of exception suggests that the models could be too conservative, but given the small sample size, they maybe just did not breach the threshold in this short window. This means that the models estimated risk levels that are higher than the realised market volatility.

Beyond the unconditional frequency of violations, we also check whether violations exhibit clustering through the IND and CC tests. The reported p_{IND} and p_{CC} values do not provide evidence against independence or conditional coverage. That said, when a model produces zero exceptions, the IND test becomes weakly informative, so these results should be interpreted with caution.

Therefore, we select the **Copula-MC**, that recorded 1 exception, resulting in a hit ratio of 4.76%, getting close to the 5% target discussed earlier. The high p-value (0.96) indicates that we strongly cannot reject the null hypothesis of correct coverage, suggesting the model is adequately calibrated for this period.

Visually, figure 10 shows the performance of the model, with only 1 violation of the 95% VaR.

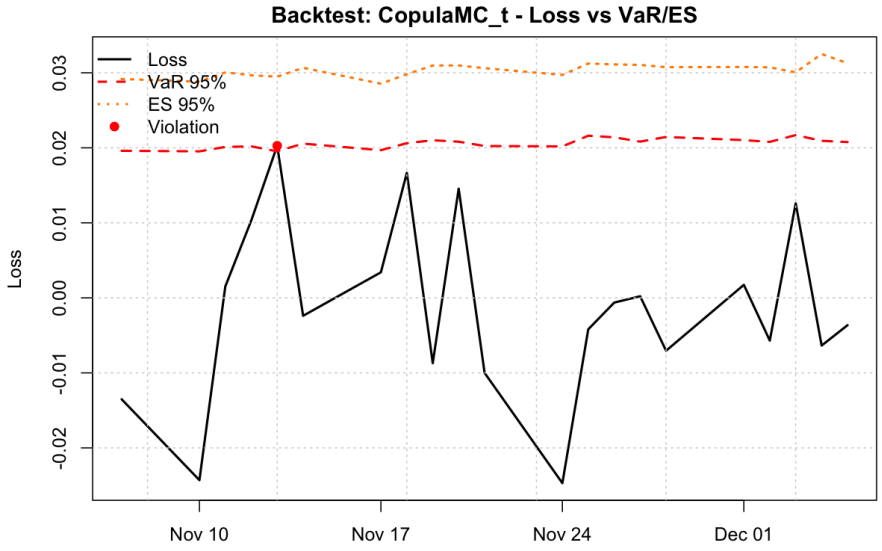


Figure 10: Realised loss with VaR and ES forecast lines

6 Conclusion

This project aimed at implementing and evaluating risk modelling techniques for our constructed equity portfolio. Our analysis of the data features confirmed the presence of two stylised facts: volatility clustering and heavy tails. Those facts were proof of non-normality of the log-returns.

By implementing a Copula-GARCH framework, we addressed the stylised facts:

- **GARCH-t** filtered the conditional volatility, accounting for the clustering observed in the Ljung-Box tests.
- **Student-t Copula** captured the symmetric tail dependence, acknowledging the non-linear risk of simultaneous asset crashes.

The backtesting results provided evidence for this approach. While benchmark models were too conservative, the Copula-GARCH Monte Carlo model achieved a great hit ratio (4.76%), and a high Kupiec p-value (0.96), showing the robustness of the model in measuring risk.